

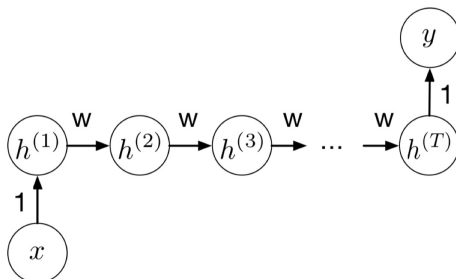
Practice Exam 2 for COMS 4776
Neural Networks and Deep Learning
Fall 2025

Name: _____

Uni: _____

1. Consider the following RNN, which has a scalar input at the first time step, makes a scalar prediction at the last time step, and uses a shifted logistic activation function:

$$f(z) = \sigma(z) - 0.5$$



- (a) Write the formula for the derivative $\overline{h_t}$ as a function of $\overline{h_{t+1}}$, for $t < T$. (Use z_t to denote the input to the activation function at time t . You may write your answer in terms of σ' , i.e., you don't need to explicitly write out the derivative of σ .)
- (b) Suppose the input to the network is $x = 0$. Notice that $h_t = 0$ for all t . Based on your answer to part (a), determine the value α such that if $w < \alpha$, the gradient vanishes, while if $w > \alpha$, the gradient explodes. You may use the fact that $\sigma'(0) = 1/4$.

2. Language Models and Transformers

- (a) In the encoder/decoder architecture for machine translation, the network reads a sentence and stores all the information in its hidden units. Describe two ways in which a bidirectional RNN encoder differs from this.

- (b) Explain an advantage of transformers over RNNs, focusing on computational efficiency. Then explain one disadvantage of transformers vs. RNNs in terms of computational efficiency.

- (c) How do *self-attention* and *cross-attention* differ?

3. The standard linear transformer variant linearizes the attention computation as follows:

$$\text{LinAtt}(Q, K, V) = D^{-1} (\phi(Q) (\phi(K)^\top V)), \quad D = \text{diag}(\phi(Q) \phi(K)^\top \mathbf{1}),$$

where $\phi(\cdot)$ is a feature map applied element-wise. Several choices of $\phi(\cdot)$ can be used, including a ReLU feature map or one that approximates standard softmax attention.

- (a) Assume that the shapes of Q , K , and V are as above. State the dimensions of $\phi(Q)$, $\phi(K)$, $\phi(K)^\top V$, and $\phi(Q) (\phi(K)^\top V)$.
- (b) Provide the computational complexity of linear attention in terms of n and d .
- (c) Provide the memory complexity of linear attention in terms of n and d .
- (d) Using your results above, explain when linear transformers would be more efficient than the standard self-attention, and why this is advantageous in practice for model training and inference.

4. Adapting Transformers

- (a) Given a scenario where you need to adapt a large language model for medical diagnosis tasks with limited computational resources, which adaptation method(s) would you recommend and why?

- (b) What is In-Context Learning (ICL)? Give an example and explain why it's considered an "inference-time adaptation" method.

5. Autoencoders

- (a) What is the training objective for a standard autoencoder?
- (b) Describe the two terms in the training objective of a VAE, and how they relate to the overall aim of the model.
- (c) Consider a trained VAE and a trained GAN. Compare how they are used to generate samples.
- (d) Specify the Autoencoder formulation that relates closely to PCA, and explain how they differ.

6. Recall the discriminator's cost function in a GAN:

$$\mathcal{J}_D = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[-\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z}}[-\log(1 - D(G(\mathbf{z})))]$$

- (a) What do $D(\mathbf{x})$ and $G(\mathbf{z})$ represent?
- (b) Explain why this is a sensible cost function for the discriminator.
- (c) One possible cost function for the generator is the opposite of the discriminator's. Explain why this is sensible.
- (d) However, in practice, the generator is usually trained with a different loss function. What cost function do we typically use for the generator?
- (e) What is the reason to use this cost function rather than the one given above?

7. The CycleGAN model

- (a) If the model is successful, what will it be able to do?
- (b) What data is required to train the model?
- (c) What might go wrong if we eliminate the discriminator terms from the cost function?
- (d) What might go wrong if we eliminate the reconstruction term from the cost function?